Lecture 30: Diode Applications: Wave Shaping Circuits

Some Other Applications: Wave-Shaping Circuits

1. **Clipping Circuits:**
   These circuits limit or remove portions of a waveform above or below a selected voltage level. They are employed in overvoltage protection, signal-amplitude limiting, noise suppression, and waveform generation.

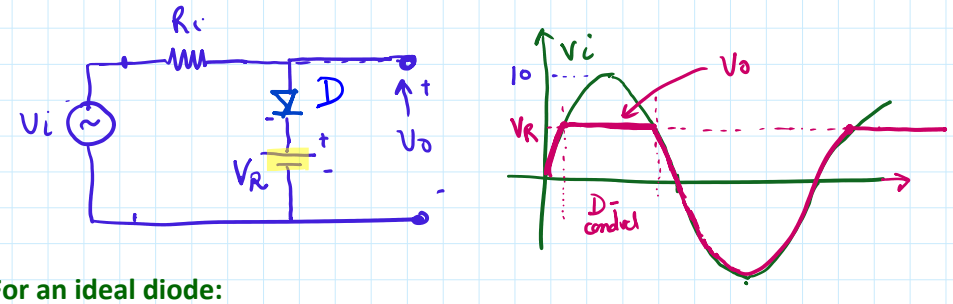
2. **Clamping Circuits:**
   These circuits shift an entire waveform to a different DC level without substantially changing its shape. They are used for DC restoration in television and communication signals, voltage-level shifting, and voltage-multiplier circuits.

3. **Comparator Circuits:**
   These circuits compare an input voltage with a reference level and indicate whether the input is higher or lower. They are used in zero-crossing detectors, overvoltage and undervoltage detectors, battery-level indicators, ADCs, and PWM circuits.

4. **Digital Logic Circuits:**
   Diodes with resistors can realize simple AND and OR logic operations. They are used in diode-logic gates, alarm circuits, signal-routing networks, keyboard matrices, and power-supply OR-ing circuits.

A. Clipping Above V_R

The circuit clips the portion of the input waveform that rises above the reference voltage V_R. Therefore, it is also called an upper-level biased clipper.



For an ideal diode:

(a) When $v_i > V_R$:
The diode is ON, and the output is limited to the reference voltage:
$v_o = V_R$

(b) When $v_i < V_R$:
The diode is OFF, and the output follows the input:
$v_o = v_i$

Therefore, the input–output relationship is
$v_o = \min(v_i, V_R)$

Thus, the output follows the input whenever $v_i < V_R$, but it remains fixed at V_R whenever the input attempts to rise above V_R.

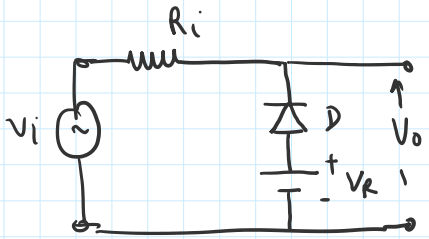
For a practical silicon diode with forward-voltage drop V_D, the diode begins conducting when

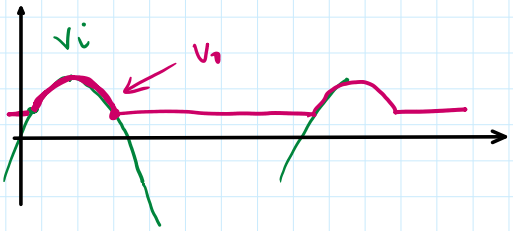
$v_i > V_R + V_D$,

and the practical clipping level is approximately
$v_o = V_R + V_D \approx V_R + 0.7\ \text{V}$.

B. Clipping Below V_R

The circuit clips the portion of the input waveform that falls below the reference voltage V_R. Therefore, it is also called a lower-level biased clipper.





For an ideal diode:

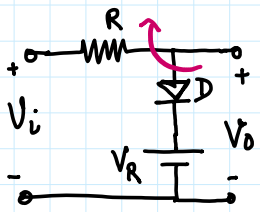
(a) When $v_i > V_R$:
The diode is OFF, and
$v_o = v_i$
(b) When $v_i < V_R$:
The diode is ON, and
$v_o = V_R$

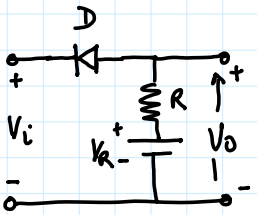
Therefore, the input–output relationship is
$v_o = \max(v_i, V_R)$

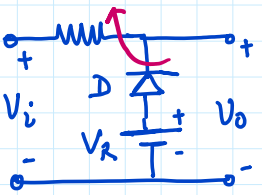
Thus, the output follows the input whenever $v_i > V_R$, but it is maintained at V_R whenever the input attempts to fall below V_R.
For a practical silicon diode with forward-voltage drop V_D, the clipping level is approximately

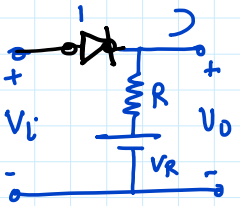
$v_o = V_R - V_D \approx V_R - 0.7\ \text{V}.$

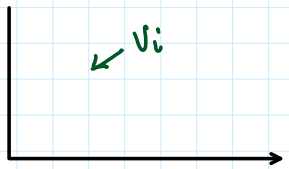
Other Similar Circuits to realize clipping:

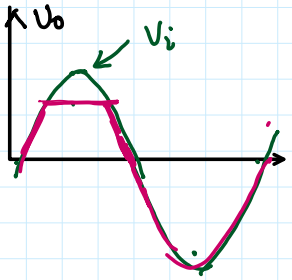


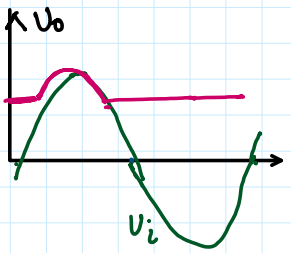


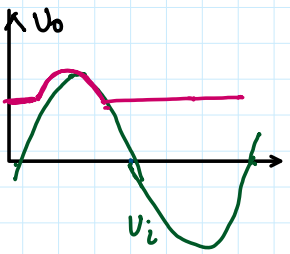




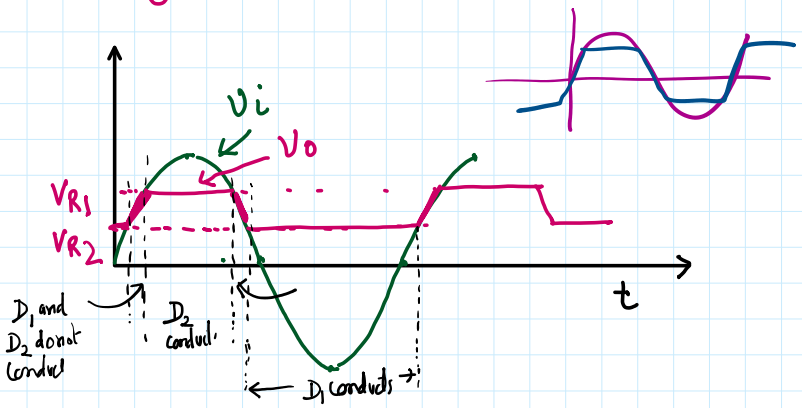








Clipping at Two Independent Levels



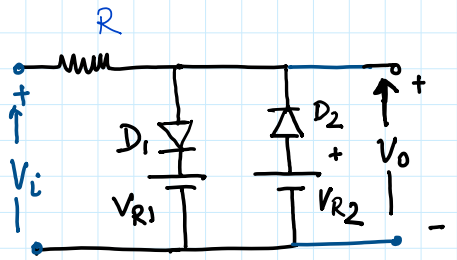
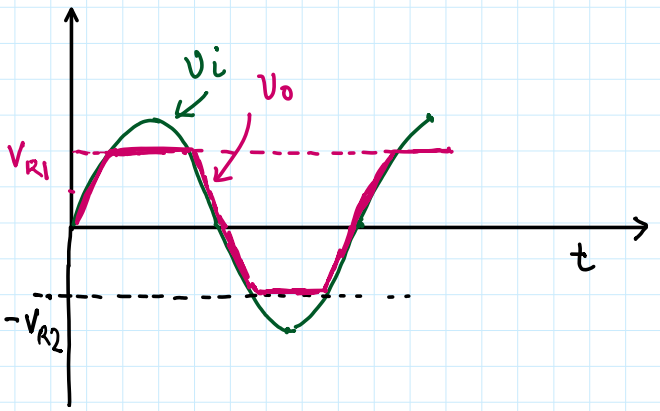
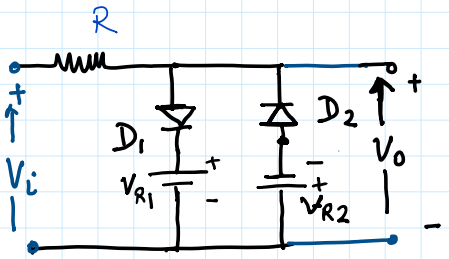


Figure: Clipping between two levels

$V_{R1} > V_{R2}$

What will be the output if V_{R1} is of opposite polarity?





Applications of Clipper Circuits: (Slicers, limiters)

1. Waveshaping in electronic circuits
2. Protection against voltage variation in transient conditions.

Peak Detector Circuit:

If R_L is very high, $\tau = R_L C$ is large

Hence $V_C = V_o = V_m e^{-t/\tau} \simeq V_m$

Diode mostly remains in non-conducting mode. Thus, half wave rectifier circuit with large values of R_L can be used as peak detector.

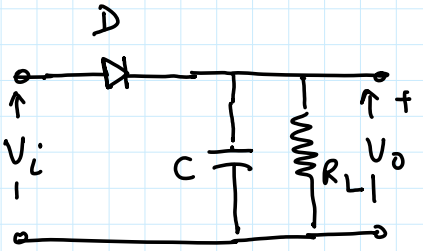
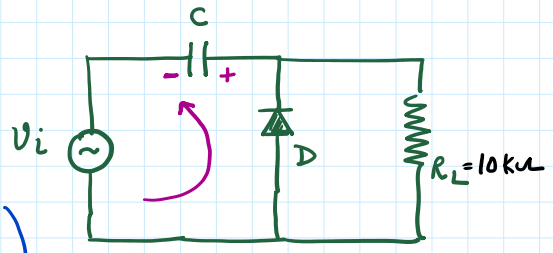


Figure: Peak detector circuit

Clamping Circuits: Clamping circuit clamp the signal at certain dc level, without changing the signal.

certain dc level, without changing the signal. These may also called as level shifters. For example consider the following circuit.

Thus mathematically $V_o = V_i + V_R$ (Shift)



* To use it as clamping circuit, the charging time of capacitor is fast, as compared to the discharging time. (This was the case with rectifier circuit with output capacitance)

Consider C = 1μF, R_f = Forward biased diode resistance = 10 Ω

Charging time constant, $\tau_c = R_f C = 10 \times 10^{-6} = 10\ \mu sec$

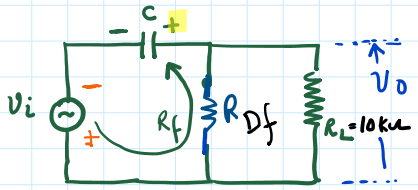
Let us say charging time is $5\tau_c = 5 \times 10 = 50\ \mu sec$

Discharging time constant $\tau_d = R_L * C = 10 \times 10^{3} \times 10^{-6} = 10\ m sec$

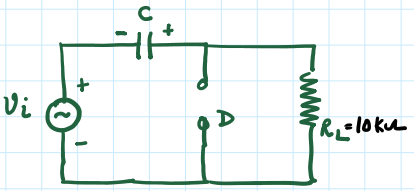
Total discharging time, $5\tau_d = 5 \times 10 = 50\ m sec.$

$t_c \ll t_d$ (This is basis of clamping circuit)

Operation: During negative half cycle, diode D is forward biased thus, diode conducts, charging capacitor C, quickly to V_{max}.



(a) During negative half cycle C is charged to V_m $V_o = 0$ during charging

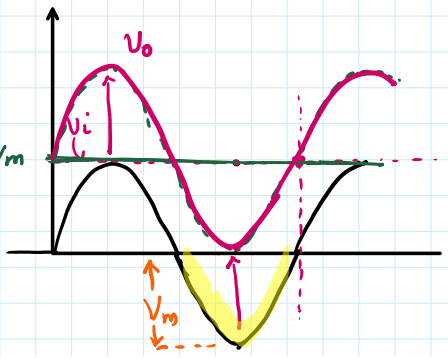


(b) During positive half cycle, diode is reverse biased, hence off,

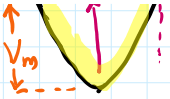
$-V_i - V_C + V_o = 0$

$V_o = V_i + V_C = V_i + V_m$ $\quad V_C = V_m$

Thus, V_o is shifted or clamped to the level of V_m, as shown in figure.



to the level of V_m, as shown in figure.



This is known as positive clamper.

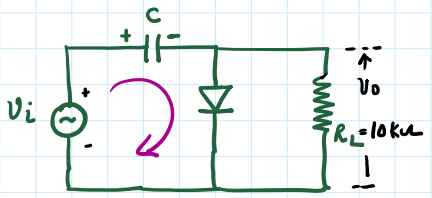
If we reverse the direction of diode pointing downward, it is called negative clamper. The output is clamped to $-V_m$.

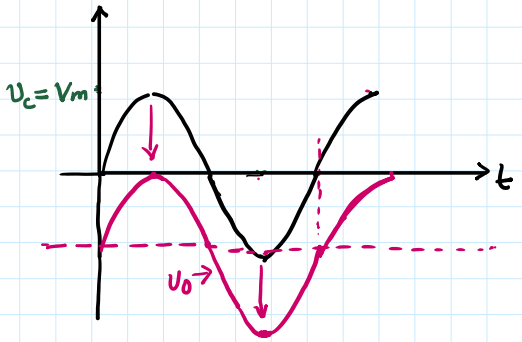
C is charged to $-V_m$ during positive half cycle, quickly with $V_0 = 0$ for a short while

$$V_i - V_C = V_0$$

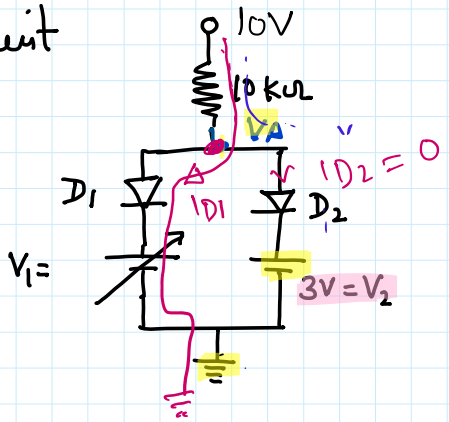
$$\Rightarrow V_0 = V_i - V_m$$

Thus, output is shifted by $-V_m$. Hence called negative clamper.





Some Examples: Consider the following circuit



a) Sketch a plot of voltage V_A versus V_1 for $0 < V_1 < 10V$.

b) Calculate current through diodes D_1, D_2 for two cases: i) $V_1 = 2V$ ii) $V_1 = 4V$

Solution: for $0 \le V_1 \le 3V$

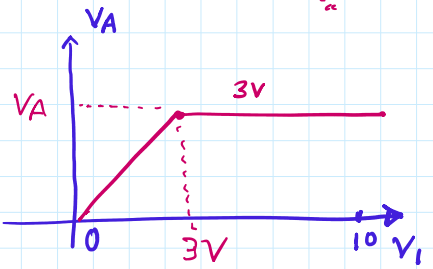
$V_A = V_1$. D_1 is ON, D_2 is OFF

for $V_1 > 3V$

D_2 is ON

D_1 is OFF

$V_0 = 3V$



(b) when $V_1 = 2V$ $\quad \frac{10-2}{\quad} = 0.8mA$

(b) When $V_1 = 2\ V$

$$I_{D1} = \frac{10-2}{10\ (k\Omega)} = 0.8\ mA$$

$$I_{D2} = 0$$

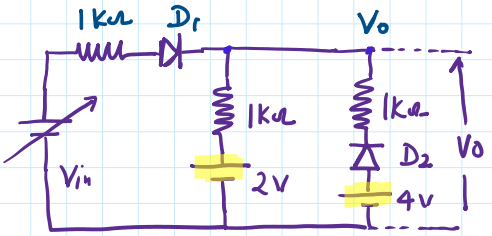
When $V_1 = 4\ V$, $V_A = 3\ V$

$$I_{D2} = \frac{10-3}{10} = 0.7\ mA$$

$$I_{D1} = 0$$

Q2 Compute V_o for

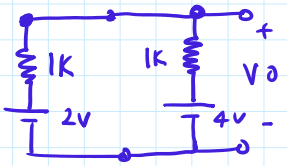
a) $V_{in} = 0V$
b) $V_{in} = 5V$
c) $V_{in} = 10V$



Solution:

a) $V_{in} = 0$

Let us assume, D_1 is OFF
D_2 is ON



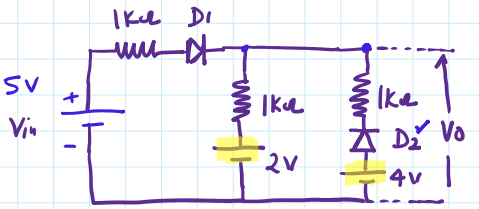
$$\frac{V_o - 2}{1} + \frac{V_o - 4}{1} = 0$$

$$\Rightarrow 2V_o = 6 \Rightarrow V_o = 3\ V$$

$V_o = 3V$ indicates that initial guess is correct as for $V_o = 3V$, D_1 is reversed biased and D_2 is forward biased.

b) When $V_{in} = 5V$

Assume { D_1 is ON, D_2 is ON



$$\frac{V_o - 5}{1} + \frac{V_o - 2}{1} + \frac{V_o - 4}{1} = 0$$

$$3V_o = 11,\ V_o = 11/3 = 3.6667$$

$V_o = 3.667\ V$ implies that D_1 forward biased and D_2 is reversed biased. Hence initial guess is reasonable. In case

and D_2 is reversed biased. Hence initial guess is reasonable. In case...

Consider D_1 ON, D_2 is OFF

$$\frac{V_0 - 5}{1} + \frac{V_0 - 2}{1} = 0$$

$$V_0 = \frac{7}{2} = 3.5\ V$$

$V_0 = 3.5\ V < 4V$

thus D_2 will conduct. Hence the assumption D_2 is OFF is incorrect.

(c) $V_{in} = 10V$, assumption, D_1 ON, D_2 OFF is more reasonable.

For this $\frac{V_0 - 10}{1} + \frac{V_0 - 2}{1} + 0 = 0$

$V_0 = 6\ V$ meets the criteria

D_1 ON and D_2 OFF.